Credit Risk and Loan Approval Analytics Platform: Final Report

Credora Finance

Project code CRDA-RISK-2024

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1. Introduction

Credora finance operates across 20 cities over 13 states and offers personal, gold, home, auto, education and business loans. Lending decisions are primarily based on single Bureau score and manual judgment which is leading to inconsistent approvals and no early warning of default.

This project helps in that particular area, wherein it builds an end-to-end decision support platform along with a joined relational database, an analytical SQL layer, loan-default prediction model, a 0-100 risk score and a Tableau dashboard.

Data Used: 4 CSV Files

- Customers: 5000
- Loan_application: 7500 (Jul 2022 to Jun 2024), of which 3,403 were disbursed.
- Credit_history: 5000
- transactions: 40079 - last 6 months

2. Objectives

- Join the 4 different datasets into one relational model with enforced keys
- Build an ML model that would predict the probability of default for each loan
- To categorize the customer in to Low/Medium/High risk categories using risk score scale 0-100.
- Create a dashboard with 8 specific components required in the PRD section 14

- Success targets ROC-AUC>= 0.80 and recall >=0.70

3. Data Overview

Customers

```
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 20 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   customer_id         5000 non-null   str
 1   first_name          5000 non-null   str
 2   last_name           5000 non-null   str
 3   gender              5000 non-null   str
 4   date_of_birth       5000 non-null   str
 5   age                 5000 non-null   int64
 6   marital_status      5000 non-null   str
 7   dependents          5000 non-null   int64
 8   education           5000 non-null   str
 9   employment_type     5000 non-null   str
 10  occupation          5000 non-null   str
 11  annual_income       5000 non-null   int64
 12  city                5000 non-null   str
 13  state               5000 non-null   str
 14  residence_type      5000 non-null   str
 15  years_at_residence  5000 non-null   int64
 16  email               5000 non-null   str
 17  phone               5000 non-null   str
 18  kyc_status          5000 non-null   str
 19  customer_since      5000 non-null   str
dtypes: int64(4), str(16)
```

- Customers: 5000 rows, 20 columns
- This is the parent dataset for the rest of the datasets available
- Primary key: “customer_id”

Credit History

```
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   credit_id                5000 non-null   str
 1   customer_id              5000 non-null   str
 2   as_of_date               5000 non-null   str
 3   credit_score             5000 non-null   int64
 4   num_open_accounts        5000 non-null   int64
 5   num_credit_inquiries_6m  5000 non-null   int64
 6   credit_utilization_ratio 5000 non-null   float64
 7   num_late_payments_30d    5000 non-null   int64
 8   num_late_payments_90d    5000 non-null   int64
 9   num_defaults_prior       5000 non-null   int64
 10  bankruptcies             5000 non-null   int64
 11  total_credit_limit       5000 non-null   int64
 12  total_outstanding_debt   5000 non-null   int64
 13  oldest_account_age_months 5000 non-null  int64
dtypes: float64(1), int64(10), str(3)
memory usage: 547.0 KB
None
```

- Credit History: 5000 rows, 14 columns
- This is the child dataset to the customer dataset
- primary key: “credit_id”

Loan Application

```
RangeIndex: 7500 entries, 0 to 7499
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   application_id           7500 non-null   str
 1   customer_id              7500 non-null   str
 2   application_date          7500 non-null   str
 3   loan_type                7500 non-null   str
 4   loan_purpose             7500 non-null   str
 5   requested_amount         7500 non-null   int64
 6   loan_term_months         7500 non-null   int64
 7   interest_rate            7500 non-null   float64
 8   applicant_monthly_income  7500 non-null   int64
 9   existing_emi             7500 non-null   int64
 10  debt_to_income_ratio     7500 non-null   float64
 11  collateral_flag          7500 non-null   int64
 12  approval_status          7500 non-null   str
 13  approved_amount          7500 non-null   int64
 14  decision_date            7327 non-null   str
 15  disbursed_flag           7500 non-null   int64
 16  loan_default             3403 non-null   float64
dtypes: float64(3), int64(7), str(7)
memory usage: 996.2 KB
```

- Loan application Dataset: 7500 rows, 17 columns
- This is the child dataset to customers dataset
- Primary key: “application_id”

Transactions

```
RangeIndex: 40079 entries, 0 to 40078
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   transaction_id    40079 non-null  str
 1   customer_id       40079 non-null  str
 2   transaction_date  40079 non-null  str
 3   transaction_type  40079 non-null  str
 4   category          40079 non-null  str
 5   channel           40079 non-null  str
 6   amount            40079 non-null  float64
 7   balance_after     40079 non-null  float64
dtypes: float64(2), str(6)
memory usage: 2.4 MB
None
```

- Transactions Dataset: 40079 rows, 8 columns
- This is the child dataset to the customer dataset and its purpose is to store the transactions done by the customers to identify the financial disciplines of the customer
- Primary key: “transaction_id”

4. Data Quality & Cleaning

Duplicate Records

Customers: checked 3 columns that should be unique.

- Customer_id: 0
- Email: 1 Email ID found to be duplicate, but it is not used for modelling and is hidden in the dashboard
- Phone: 0

Credit History: It has 2 columns with unique values and none of them have duplicates in them

- Credit_id: 0
- Customer_id: 0

Loan Applications: application_id is unique (no duplicates)

- Application_id: 0
- customer_id: It repeats as expected as a single customer may have provided multiple loan applications

Transactions: Transaction ids are unique

- Transaction_id: 0
- customer_id: It repeats as expected as a single customer may have multiple transactions

Invalid Data

The only issue was that the date column were stored as text, these were converted dates during data cleaning

Missing Values

- No missing values in Customers, Credit History or Transactions.
- Loan Applications: decision_date is blank for 173 applications still Under Review.
- Loan Applications: loan_default is blank for 4,097 applications that were not disbursed. This is expected, because only the 3,403 disbursed loans can default, and only those are used for modelling.

Summary Stats

Customers

	age	dependents	annual_income	years_at_residence
count	5000.000000	5000.000000	5.000000e+03	5000.00000
mean	38.388400	1.319200	5.795758e+05	9.94840
std	10.560277	1.135063	4.225968e+05	6.52842
min	21.000000	0.000000	1.200000e+05	0.00000
25%	30.000000	0.000000	2.941750e+05	4.00000
50%	38.000000	1.000000	4.748000e+05	9.00000
75%	46.000000	2.000000	7.280500e+05	15.00000
max	74.000000	6.000000	4.636900e+06	24.00000

Observation: Annual income shows a wide range, suggesting potential high-income outliers which are examined in the income histogram.

Credit History

1

	credit_score	num_open_accounts	num_credit_inquiries_6m	...	total_credit_limit	total_outstanding_debt	oldest_account_age_months
count	5000.00000	5000.000000	5000.000000	...	5.000000e+03	5.000000e+03	5000.000000
mean	648.93440	3.490400	1.670400	...	3.193644e+05	1.095843e+05	192.691600
std	85.40396	1.848294	1.326694	...	2.769357e+05	1.116656e+05	122.504552
min	300.00000	0.000000	0.000000	...	2.500000e+04	2.500000e+03	6.000000
25%	595.00000	2.000000	1.000000	...	1.380000e+05	4.040000e+04	96.000000
50%	652.00000	3.000000	1.000000	...	2.435000e+05	7.530000e+04	190.000000
75%	706.00000	5.000000	2.000000	...	4.070000e+05	1.396000e+05	281.000000
max	900.00000	14.000000	8.000000	...	3.651000e+06	1.682000e+06	480.000000

[8 rows x 11 columns]

Observation: Credit scores range from 300 to 900 (average about 649). Credit limit and outstanding debt are right-skewed, with a few very large values.

Loan Applications

1

	requested_amount	loan_term_months	interest_rate	applicant_monthly_income	...	collateral_flag	approved_amount	disbursed_flag	loan_default
count	7.500000e+03	7500.00000	7500.000000	7500.000000	...	7500.000000	7.500000e+03	7500.000000	3403.000000
mean	2.368134e+06	50.51680	13.744532	48526.733333	...	0.521200	5.883009e+05	0.453733	0.234205
std	2.698808e+06	40.37292	4.124718	35707.059254	...	0.499584	1.288144e+06	0.497888	0.423563
min	3.100000e+04	12.00000	8.010000	10000.000000	...	0.000000	0.000000e+00	0.000000	0.000000
25%	6.200000e+05	24.00000	10.290000	25275.000000	...	0.000000	0.000000e+00	0.000000	0.000000
50%	1.288000e+06	36.00000	12.740000	39900.000000	...	1.000000	0.000000e+00	0.000000	0.000000
75%	2.873750e+06	60.00000	16.590000	60400.000000	...	1.000000	6.450000e+05	1.000000	0.000000
max	1.200000e+07	240.00000	24.000000	386400.000000	...	1.000000	1.171600e+07	1.000000	1.000000

[8 rows x 10 columns]

Observation:

- Requested loan amounts also span a large range are examined using distribution charts.
- Approved amount contains zero values, this was checked: zero means the application was Rejected or Under Review.
- The loan_default column contains a substantial number of missing values (4,097 of 7,500 records), these were excluded from modelling, as only disbursed loans have an outcome

Transactions

```
1
              amount  balance_after
count    40079.000000   4.007900e+04
mean     14040.707134   8.000340e+04
std      18835.280337   7.786162e+04
min        100.390000   1.000000e+02
25%       3566.420000   2.848772e+04
50%       7928.110000   5.757170e+04
75%      16313.230000   1.051002e+05
max     359766.320000   1.268816e+06
```

Observation: Transaction amounts range from ₹100 to about ₹3.6 lakh (median about ₹7,900). Both amount and balance are right-skewed.

Data Cleaning

Customer Dataset

- Gender converted to numbers Female= 0, Male= 1
- DOB and customer since converted to DateTime
- Marital status, employment type, residence type converted to category type
- KYC Status converted to Pending = 0, Verified= 1
- Spaces removed from occupation and state; dashes removed from phone numbers

Credit History Dataset

- The datatype of “as_of_date” column has been changed to Datetime
- Logical consistency has been checked. The credit_utilization_ratio was compared with total_outstanding_debt / total_credit_limit. The maximum absolute difference was 0.002, which is consistent with rounding/floating-point precision and does not indicate inconsistent records.

Transaction Dataset

Logical checks:

- Salary, Refund, and Transfer In row are in transaction_type = Credit and rest of the categories are in debit.
- Transaction type converted to numbers Debit= 0, Credit= 1
- Transaction date converted to date type
- channel and category converted to category type

Loan Applications

- application_date and decision_date converted to date type
- Spaces removed from loan_type, loan_purpose and approval_status, then set to category type
- Logical check: every decision_date is on or after its application_date (no violations)
- Leakage columns (approval_status, approved_amount, decision_date, disbursed_flag) identified for exclusion from modelling

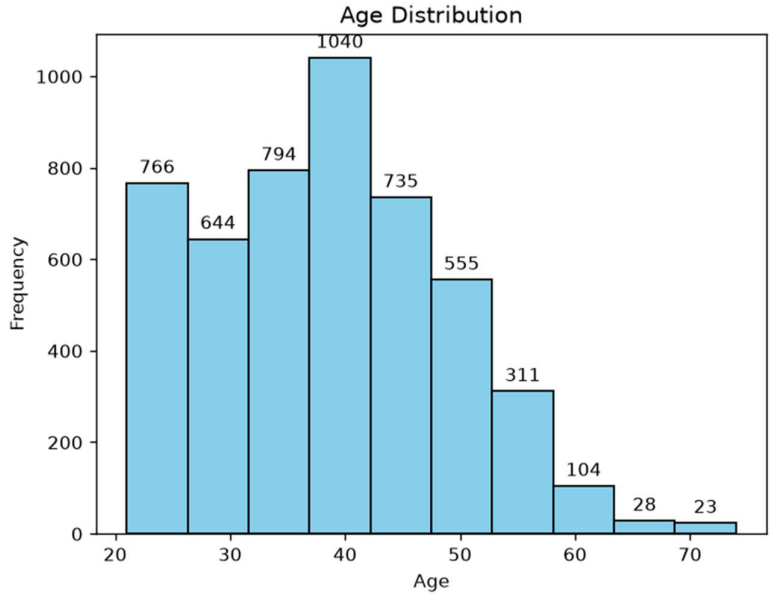
Cleaned files were saved as cleaned_customers.csv, cleaned_credit_history.csv, cleaned_loan_applications.csv and cleaned_transactions.csv.

5. Exploratory Data Analysis

Univariate Analysis

Customer Dataset

Age Distribution Histogram

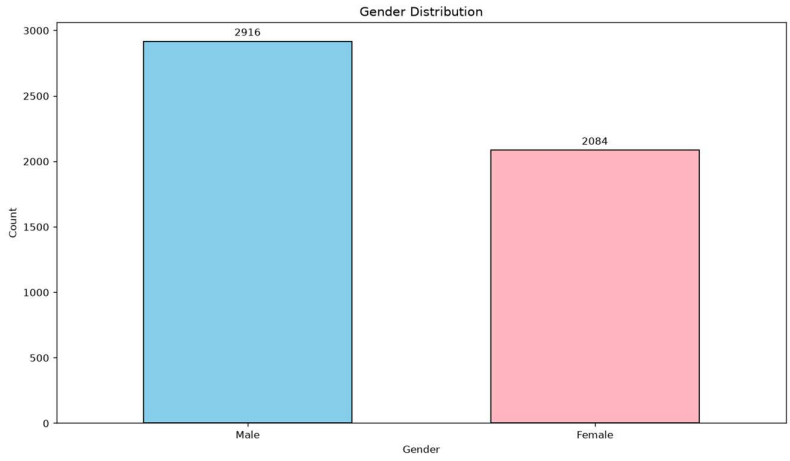


Observation:

- The data is right skewed
- The age category of population is between 21 and 74
- The highest age category of people taking credit is between 30-45

- The 2nd highest age category of people taking credit is between 20-25
- The customers beyond the age 50 taking credit is declining.
- The people who are into taking credit are largely the working-class people between the age 20-50.

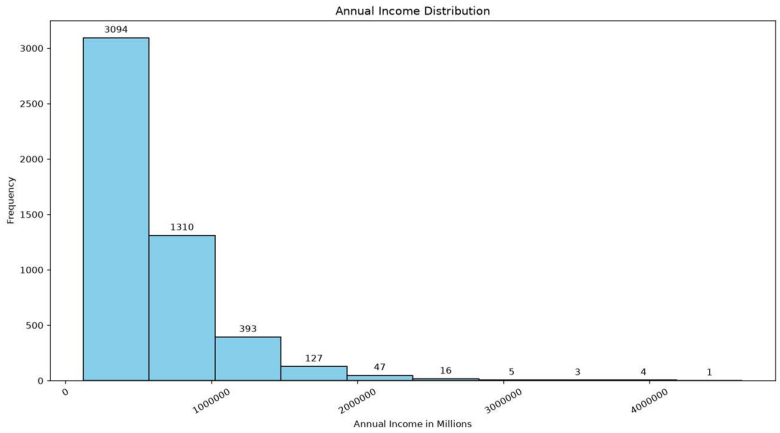
Gender Distribution



Observation:

- The total no. of sample is 5000
- The no. of males taking credit is higher in comparison to the females taking the credit
- Males are 58.3% and females are 41.7% of the total sample.
- The difference between male to female is 832 with male being higher creating a gender imbalance

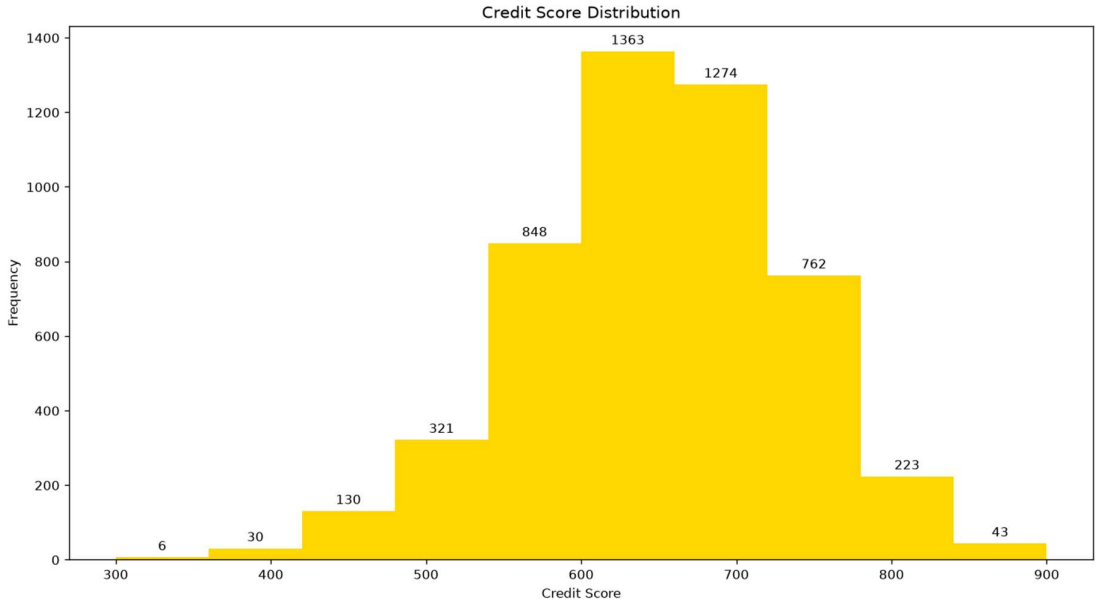
Annual Income Distribution Histogram



Observation:

- The dataset is right skewed
- Most of the customers fall below the salary bracket of approx. Rs. 500,000 which is where the peak is with 3094 customers.
- As the annual income is increasing the number of people taking credit is decreasing
- Only 1 person amongst the sample has taken credit who is earning more than 4 million in annual income
- Most credit takers belong to lower- and middle-income groups

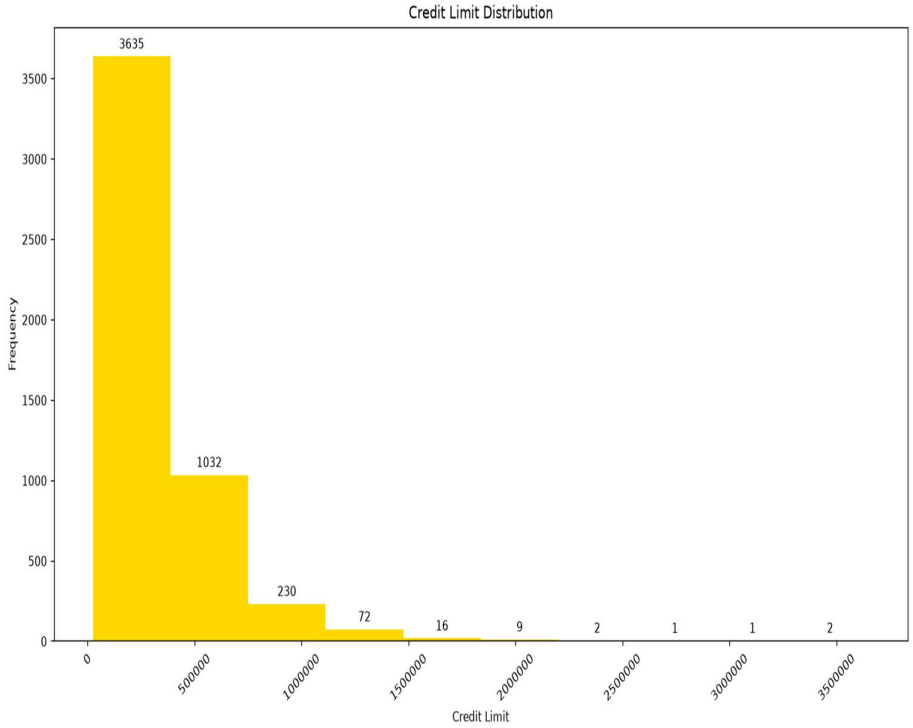
Credit Score Distribution Histogram



Observation:

- The highest number of customers have a credit score between 600-700
- The customers with credits score more than 800 are scant
- The data is clustered between 550-750

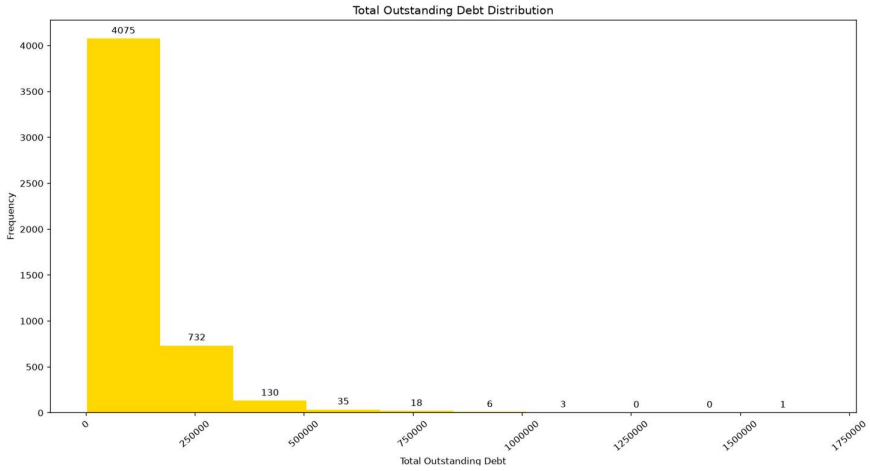
Total credit limit distribution Histogram



Observations:

- The data is positively skewed
- Majority of the customers (3,635) have a credit limit below approx. Rs. 4 lakh.
- As the credit limit increases the number of customers decreases sharply.
- People who have a credit limit more than 1.5 million is very rare in this data set.
- Most customers in this dataset have been given relatively low credit
- Only 31 customers have a credit limit above 1.5 million.

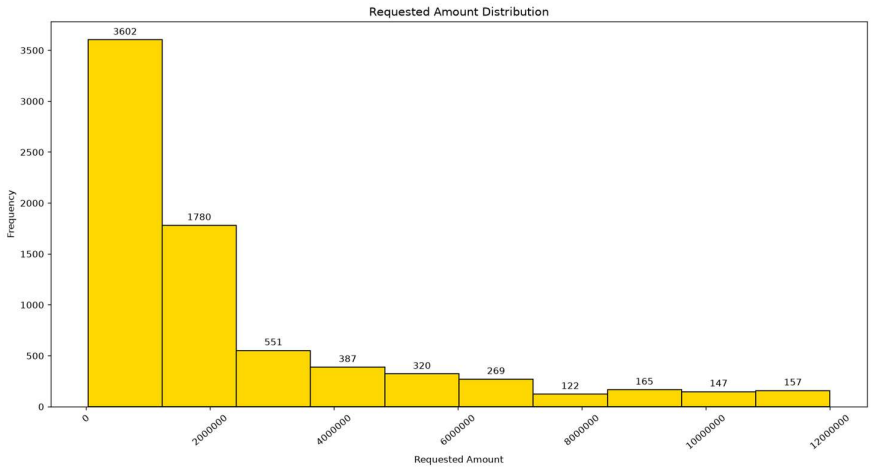
Total Outstanding Debt



Observation:

- The data is positively skewed
- 4075 customers out of 5000 have outstanding debt below approx. Rs. 1.7 lakh.
- The number of customer decreases sharply as the outstanding debt increases.
- Customers with debts above approx. Rs. 3.4 lakh are rare.
- This chart also represents that the customers refer lesser outstanding debts.

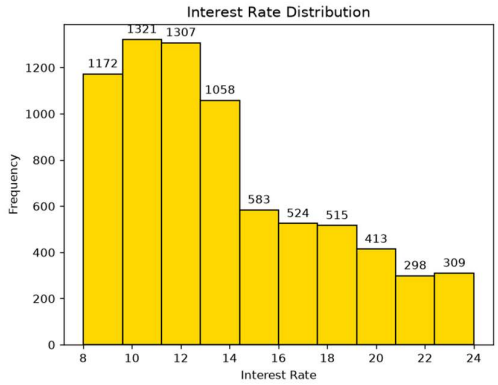
Requested Amount Distribution



Observations:

- The data is highly right skewed
- Nearly half of the applications (3,602, 48%) requested below approx. Rs. 12 lakh
- The credit amount request drops sharply above approx. Rs. 24 lakh
- Approx. 28% of applications have requested credit above approx. Rs. 24 lakh
- Most of the applications are requesting lower to moderate credit amount.

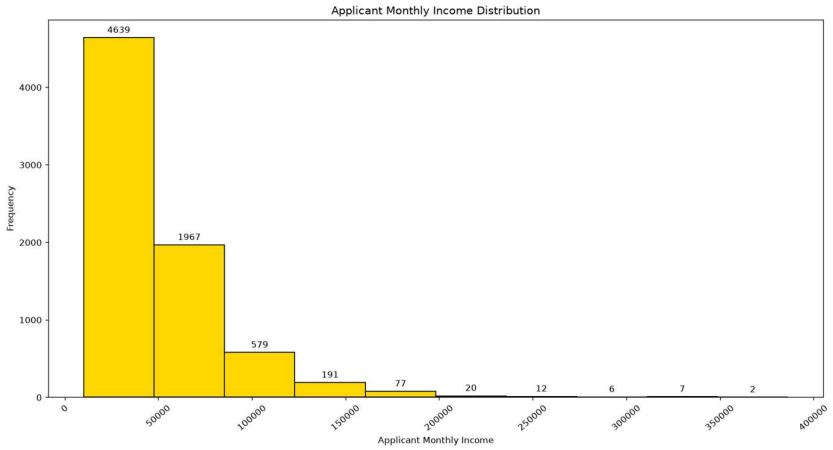
Interest Rate



Observation:

- The interest rate distribution is slightly right skewed
- Most applications have interest rate between 8% and 14% (about 65% of applications).
- The highest frequency of interest rates around 10%-12%
- The frequency decreases as the interest rates increase.

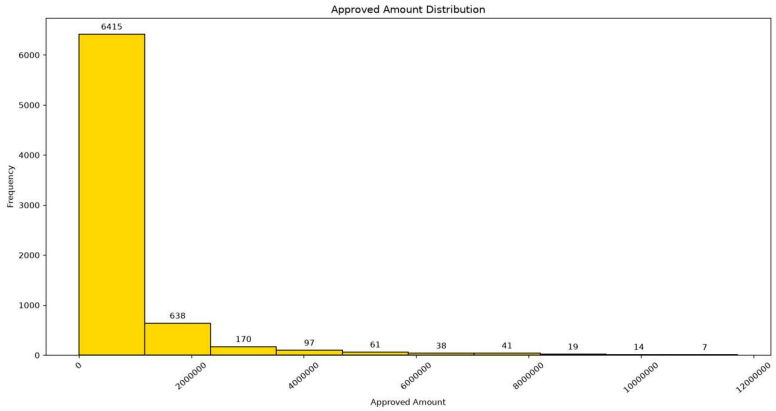
Applicant Monthly Income



Observation:

- The data is highly skewed right.
- 4,639 applications have a monthly income under **approx.** Rs. 50,000.
- Applications with a monthly income above approx. Rs. 1.2 lakh are rare (about 315).
- The monthly income of customers drops sharply after Rs. 100,000
- Most applications fall in lower to middle income group.

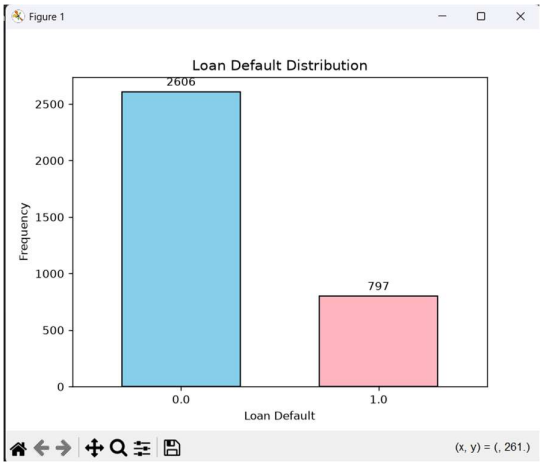
Approved Amount



Observation:

- The dataset is highly skewed right. The first bar includes 4,097 zero values: applications that were Rejected or Under Review, so nothing was approved.
- Most approved amounts fall in the first bar, up to approx. Rs. 12 lakh.
- The approved amount drops sharply after approx. Rs. 12 lakh
- Approved amounts above approx. Rs. 23 lakh are uncommon (about 447 applications).
- Most of the first bar is zero values (4,097 applications that were Rejected or Under Review). Among approved loans, most amounts are below approx. Rs. 12 lakh.

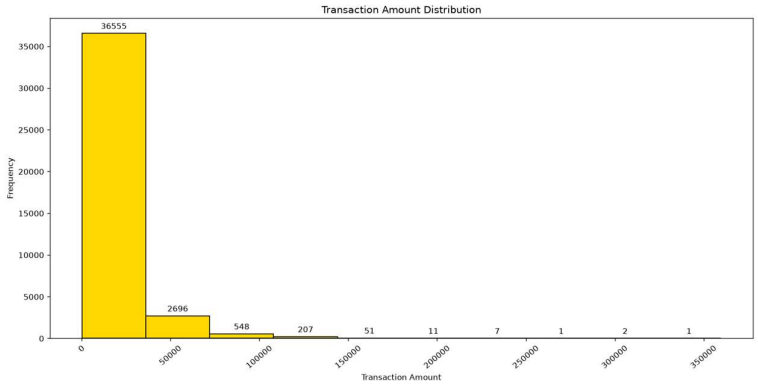
Loan Default



Observation:

- 4097 cells in this column are missing because those applications were never disbursed.
- This is a binary category column with 0 no default and 1 for defaulted.
- Of the 3,403 disbursed loans, 2,606 were repaid (76.6%) and 797 defaulted (23.4%).
- Most loans are repaid, but a 23.4% default rate is high, so predicting default is important.

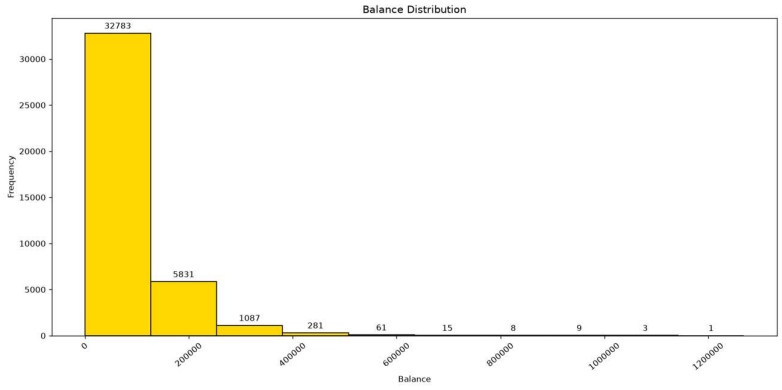
Transaction Amount



Observation:

- The Transaction Amount distribution is extremely right-skewed.
- The vast majority of transactions (36,555) are below approx. **₹36,000**.
- Frequency drops sharply after approx. **₹36,000**.
- Transactions above ₹100,000 are rare.
- Extremely high transaction amounts (above ₹200,000) are almost negligible.
- Most transactions in this dataset are of low value.

Balance After Transaction

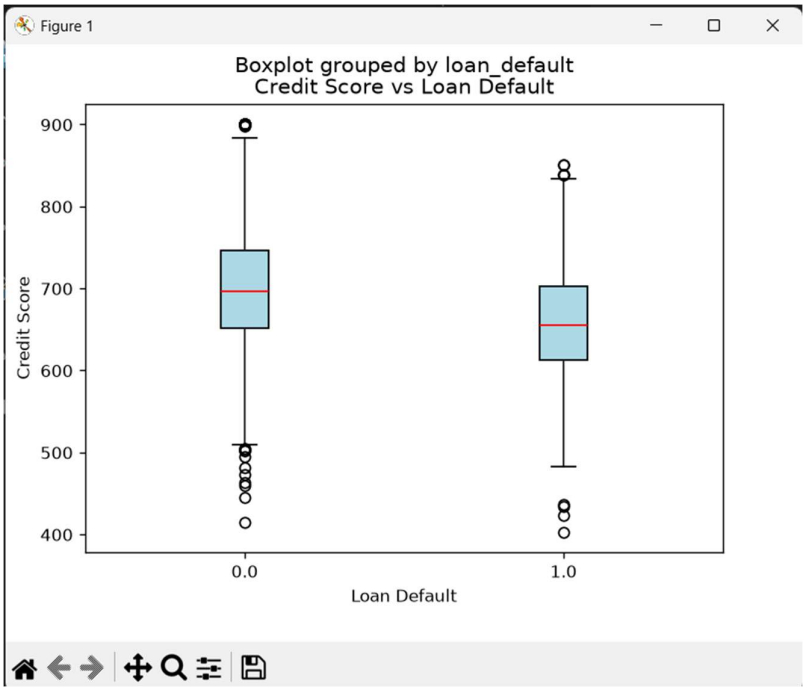


Observation:

- The Balance distribution is highly right-skewed.
- The majority of transactions (32,783) have a balance below ₹200,000.
- Frequency drops sharply as the balance increases.
- Balances above ₹400,000 are rare.
- Extremely high balances (above ₹800,000) are almost negligible.
- Most customers maintain relatively low account balances.

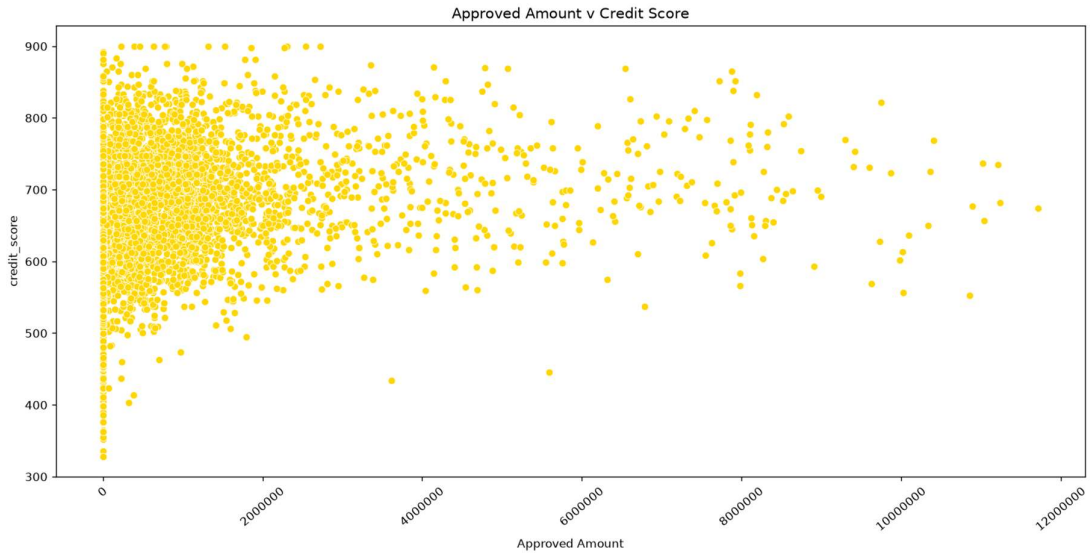
Bi-variate Analysis

Box Plot for Credit Score v Loan Default



- Only the 3,403 disbursed loans have a default outcome; the 4,097 blank values are excluded from this chart.
- This boxplot compares Credit Score between non-defaulters (0) and defaulters (1).
- Non-defaulters (0) have a higher median credit score (around 700) compared to defaulters (around 655).
- The overall credit score distribution for non-defaulters is slightly higher than that of defaulters.
- Both groups show some outliers (very high and very low credit scores).
- There is a visible difference in credit scores between the two groups, but the overlap is still significant.
- Customers with higher credit scores are less likely to default.

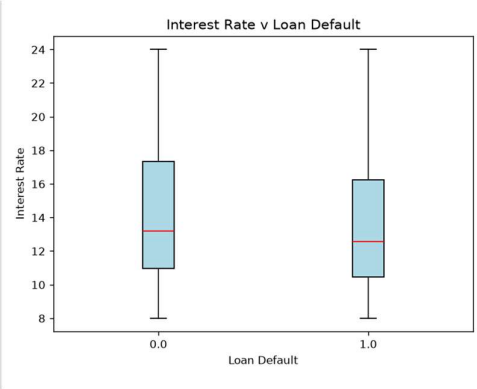
Credit Score & Approved Amount Scatter Plot



Observation:

- The analysis shows the relationship between the amount approved by the NBFC to applicant's credit score.
- The scatter plot suggests that a large cluster of amounts approved are under Rs. 2 million and most of the customers have the credit score between 600-800
- Most data points are concentrated in the lower approved amount range.
- Customers who were approved for more than Rs. 4 million are very few
- Customers who were approved for more than Rs. 4 million most of them have a credit score between 600-800
- The vertical line of points at 0 are applications that were not approved (Rejected or Under Review)
- Customers with very low credit scores (below about 500) appear almost only at 0, so low-score applications were mostly not approved.

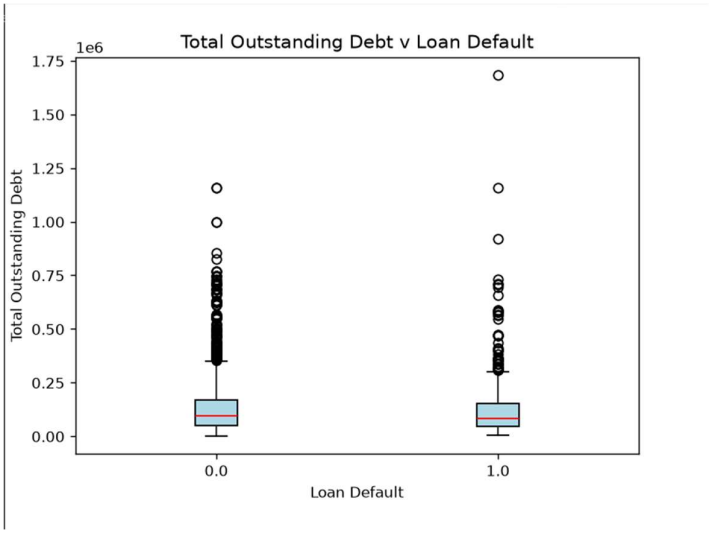
Interest Rate × Loan Default



Observation:

- The boxplot compares Interest Rate between non-defaulters (0) and defaulters (1).
- The medians are close: about 13% for non-defaulters and about 12.5% for defaulters.
- The overall distribution (box and whiskers) of interest rates is very similar for both defaulters and non-defaulters.
- There are no outliers present

Outstanding Debt × Loan Default Box Plot

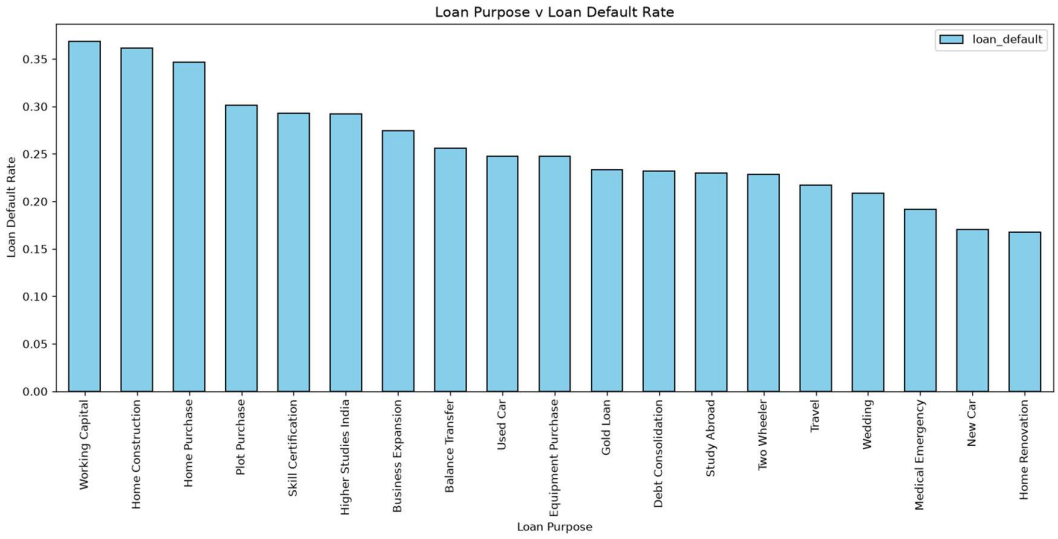


Observation:

- This boxplot compares Total Outstanding Debt between non-defaulters (0) and defaulters (1).
- The median outstanding debt is low and almost the same for both groups.
- Most customers in both groups have relatively low outstanding debt.

- Both groups show a large number of outliers with high outstanding debt.
- Non-defaulters show more high-debt outliers, but the single highest debt is a defaulter.

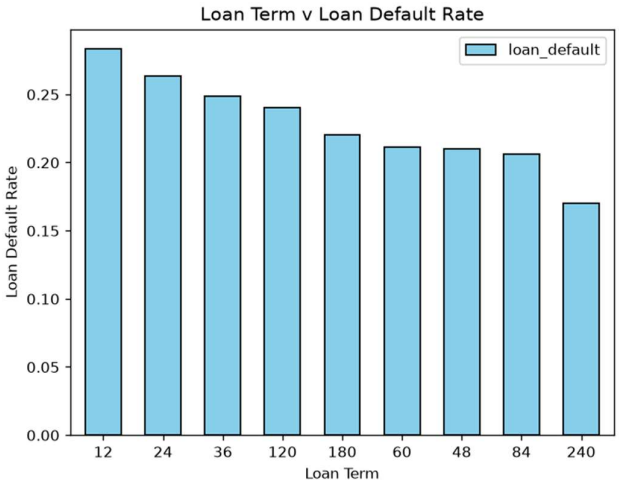
Loan Purpose × Loan Default Bar Chart — default rate



Observations:

- This bar chart shows the default rate for different loan purposes.
- Highest default rates
    - Working Capital approx. 37%
    - Home Construction approx. 36%
    - Home Purchase approx. 35%
- Lowest default rates
    - Home Renovation approx. 17%
    - New Car approx. 17%
    - Medical Emergency approx. 19%
- Loans taken for business-related and home-related purposes (Working Capital, Home Construction, Home Purchase) have higher default rates.
- Loans like Wedding, Travel, New Car, and Home Renovation have relatively lower default rates.
- Loan Purpose appears to have a noticeable relationship with the likelihood of default.

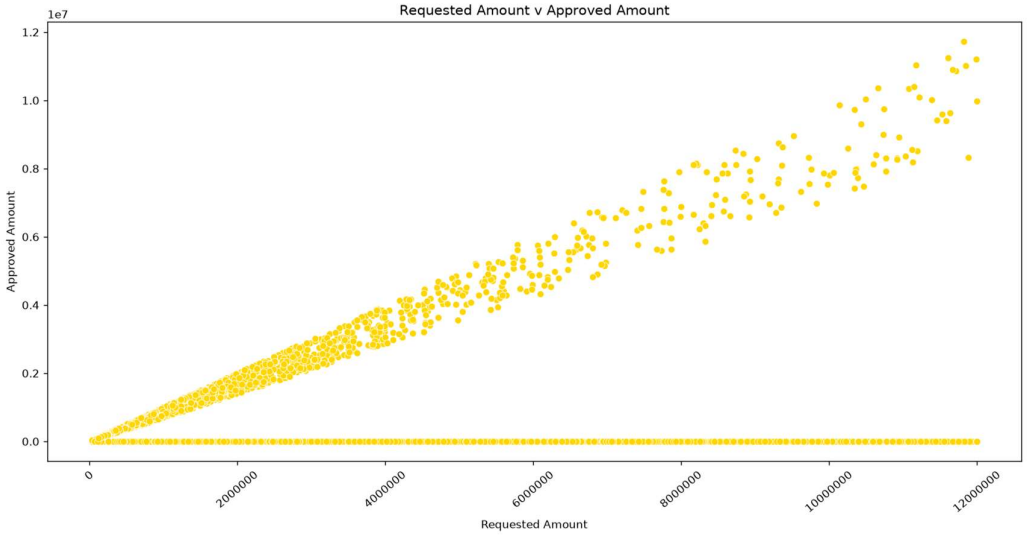
Loan Term × Loan Default Bar Chart — default rate



Observations:

- This bar chart shows the default rate across different loan terms (in months).
- Highest default rate: 12-month loans approx. 28%
- Lowest default rate: 240-month loans approx. 17%
- Shorter loan terms (12, 24, 36 months) generally have higher default rates.
- Longer loan terms (especially 240 months) have lower default rates.
- The trend is not steady: 120 and 180-month loans have higher default rates than 48, 60 and 84-month loans.

Requested Amount × Approved Amount

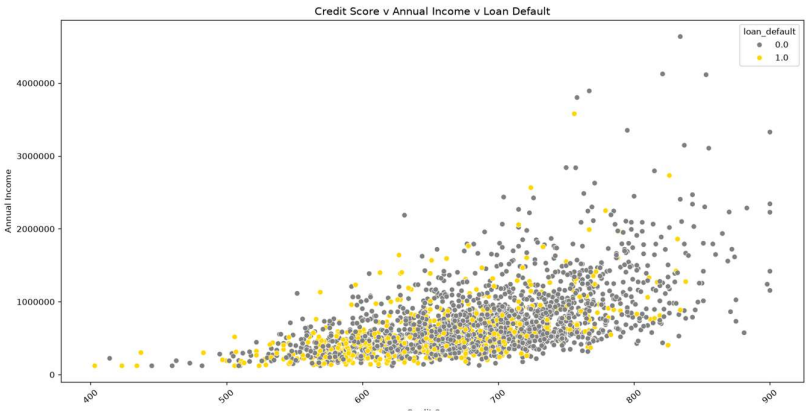


Observations

- This scatter plot shows the relationship between Requested Amount and Approved Amount.
- There is a clear positive linear relationship — as the requested amount increases, the approved amount also tends to increase.
- Many applications are approved for an amount close to what was requested (points lie near the diagonal line).
- A large number of points are present along the bottom (Approved Amount ≈ 0), indicating that some applications received **no** approval (Rejected or Under Review).
- Higher requested amounts generally receive higher approved amounts, but not always the full requested value.

Multivariate Analysis

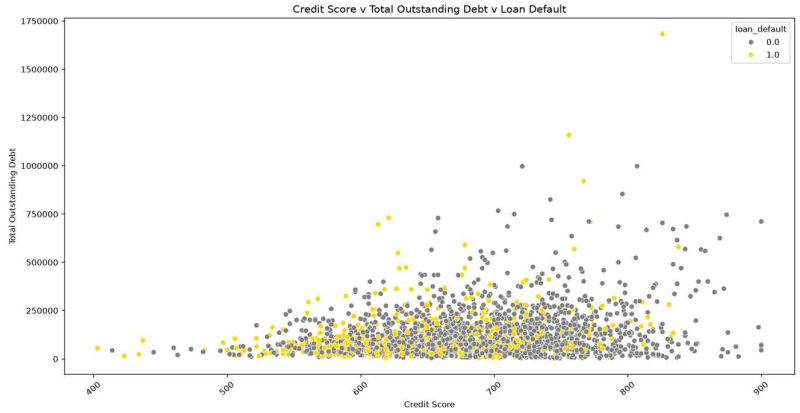
Credit Score × Annual Income × Loan Default: Scatter plot, differentiated by default status



Observations

- This scatter plot shows the relationship between Credit Score and Annual Income, with points coloured by Loan Default status (Gray = No Default, Yellow = Default).
- There is a positive relationship between Credit Score and Annual Income — higher credit scores are generally associated with higher annual income.
- Most customers are concentrated between Credit Score 600–800 and Annual Income below Rs. 1,500,000.
- Defaulters (yellow points) are present across different credit scores and income levels. Defaulters appear more frequently in the lower and middle ranges of both Credit Score and Annual Income.
- The top-right region (high Credit Score + high Annual Income) has mostly non-defaulters (gray points).

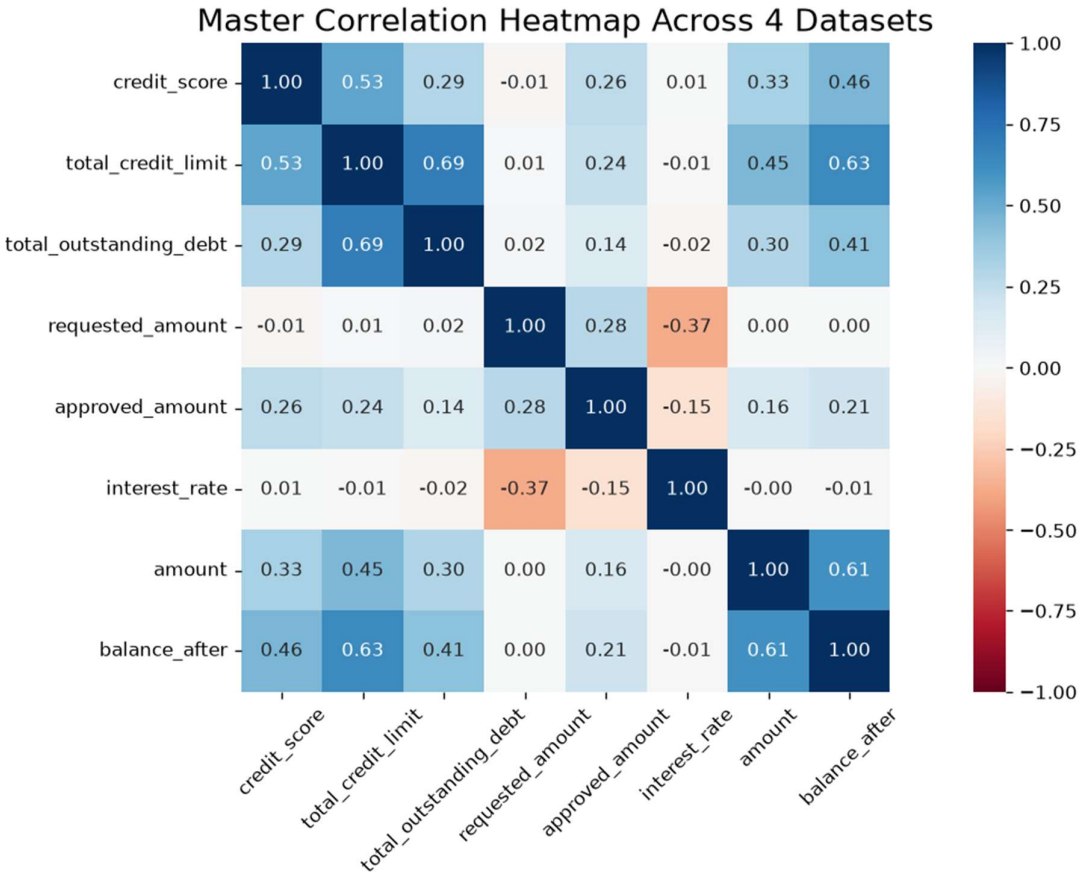
Credit Score × Outstanding Debt × Loan Default



Observations:

- This scatter plot shows the relationship between Credit Score and Total Outstanding Debt, coloured by Loan Default (Gray = No Default, Yellow = Default).
- Most customers have Total Outstanding Debt below Rs. 500,000, regardless of credit score. There is a weak positive trend — higher credit scores tend to have slightly higher outstanding debt, but the relationship is not strong.
- Defaulters (yellow points) are spread across different credit score levels.
- A large number of defaulters are concentrated in the lower outstanding debt region (below Rs. 500,000).
- High outstanding debt cases (above Rs. 750,000) are relatively few and include both defaulters and non-defaulters.

Correlation Heatmap



Observations:

- This heatmap shows the correlation between numerical variables across the datasets.
- Strong Positive Correlations:
  - total_credit_limit & total_outstanding_debt → 0.69
  - total_credit_limit & balance_after → 0.63
  - amount & balance_after → 0.61
  - credit_score & total_credit_limit → 0.53
- Moderate Positive Correlations:
  - credit_score & amount → 0.33
  - total_credit_limit & amount → 0.45
  - requested_amount & approved_amount → 0.28
  - credit_score & approved_amount → 0.26
  - credit_score & balance_after → 0.46
- Negative Correlations:
  - requested_amount & interest_rate → -0.37
  - approved_amount & interest_rate → -0.15

- Weak / No Correlation:
    - interest_rate has almost no correlation with credit_score, credit_limit, outstanding debt, or balance.
    - requested_amount has almost no correlation with credit_score or credit_limit.

EDA Summary

Customer Dataset

Age Distribution:

- The data is right skewed
- Most people taking credit are between the age 20-50
- The data is largely concentrated around working age between 30-45

Gender distribution:

- Males are 58.3% and females are 41.7% of the total sample.
- The difference between male to female is 832 with male being higher creating a gender imbalance in our dataset

Annual Income distribution:

- Highly right skewed
- Majority of the people taking credit earn below 500,000
- Most credit customers belong to lower income groups and high-income individuals are very rare in this dataset

Credit History Dataset

Credit Score:

- Most customers have a credit score between 600-700; scores above 800 are scant.

Credit Limit:

- The data is highly right skewed
- Most of the customers have credit limit below 400,000

- Most of the customers are low to moderate credit limits and customer with higher credit limits are very rare.

Total Outstanding Debt

- The data distribution is right skewed
- 4075 customers have outstanding debt below 200,000
- Majority of the customers are maintaining a low outstanding debt. High outstanding debts are very rare

Requested Amount

- The data is highly right skewed.
- Nearly half of the applications are below approx. Rs. 12 lakh
- Most customers are applying for lower to moderate credit amounts, while higher requested amounts form a smaller portion of the data.

Interest Rate

- The interest rate distribution is slightly right skewed.
- Most customers have interest rate between 8% and 14%.
- Most customers are charged moderate interest rates between 10% and 12%.

Applicant Monthly Income

- The data is highly skewed right.
- 4639 customers have their monthly income fall under Rs. 50,000.
- Most of the customers fall in lower to middle income group and customers with higher monthly income are rare.

Approved Amount

- The dataset is highly skewed right
- Many values are zero (applications Rejected or Under Review).
- Most approved amounts are below approx. Rs. 12 lakh
- Most customers received low to moderate approved amounts. High approved amounts are very uncommon.

Loan Default

- 2606 of the disbursed loans in the total available data have not defaulted which is 76.6%
- 797 of the disbursed loans in the total available data have defaulted which is 23.4%
- Majority of customers have paid off their loans. The default rate of 23.4% is high

Transaction Amount

- Extremely right-skewed distribution.
- Majority of transactions are below ₹50,000.
- Most transactions are of low value. High-value transactions are very uncommon.

Balance After Transaction

- Highly right-skewed distribution.
- Majority of transactions have balances below ₹200,000.
- Most customers maintain low account balances. High balances are very uncommon in this dataset.

Credit Score V Loan Default

- Non-defaulters have a higher median credit score than defaulters.
- Higher credit scores are associated with lower chances of loan default. Credit Score appears to be a useful indicator for predicting default risk.

Approved Amount V Credit Score

- Most approved amounts are below ₹2,000,000 with credit scores between 600–800.
- There is no strong linear relationship between Credit Score and Approved Amount. High approved amounts are rare regardless of credit score.

Interest Rate vs Loan Default

- Median interest rates are almost the same for both defaulters and non-defaulters.
- Interest Rate does not appear to have a strong relationship with loan default in this dataset.

Total Outstanding Debt vs Loan Default

- Median outstanding debt is similar for both defaulters and non-defaulters.
- Total Outstanding Debt does not show a strong difference between defaulters and non-defaulters. It may not be a strong standalone predictor of loan default.

Loan Purpose vs Loan Default Rate

- Working Capital, Home Construction, and Home Purchase have the highest default rates.
- Home Renovation and New Car have the lowest default rates.
- Loan Purpose is an important factor in default risk. Business and certain home-related loans show higher default rates compared to personal loans.

Loan Term vs Loan Default Rate

- Shorter loan terms have higher default rates.
- Longest term (240 months) has the lowest default rate.

Requested Amount vs Approved Amount

- Strong positive relationship between Requested Amount and Approved Amount.
- Higher requested amounts generally lead to higher approved amounts. However, a significant number of applications receive zero approval.

Credit Score vs Annual Income vs Loan Default

- Positive relationship exists between Credit Score and Annual Income.
- Customers with higher Credit Score and higher Annual Income are less likely to default. Default risk is higher in the lower-to-middle ranges of both variables.

Credit Score vs Total Outstanding Debt vs Loan Default

- Most customers have low outstanding debt (below Rs. 500,000).
- There is no strong clear pattern separating defaulters and non-defaulters based on Credit Score and Outstanding Debt together. Default cases are present across different levels of both variables.

Correlation Heatmap

- Strongest relationships: Credit Limit with Outstanding Debt (0.69) and Balance (0.63).
- Credit Score is moderately correlated with Credit Limit and Balance.
- Interest Rate has a negative correlation with Requested Amount (-0.37).
- Credit Limit, Outstanding Debt, and Balance are closely related. Interest Rate shows little relationship with most other variables except Requested Amount.

EDA Conclusion

The exploratory analysis, performed on the raw data, identified several factors associated with credit risk and loan performance. Credit Score demonstrated a meaningful relationship with loan default, while Loan Purpose and Loan Term showed notable differences in default rates. Most financial variables exhibited right-skewed distributions, indicating the presence of a small number of high-value observations. Correlation analysis revealed moderate to strong relationships among Credit Limit, Outstanding Debt, and Account Balance. The findings from EDA provide a foundation for the subsequent Data Cleaning and Feature Engineering stages.

6. Database and SQL Insight

Inserted all the 4 raw datasets to PostgreSQL with primary and foreign keys. The UNIQUE constraint on email was removed, because one email appears for two customers whose other details are all different (so they are two genuine customers, not a duplicate record). All other constraints were kept, and checks showed 0 orphan foreign keys. The 10 analytical queries from PRD 12.4 were run; the results are below.

Credit Score Analysis

	credit_band text	customers bigint	avg_score numeric	avg_utilization numeric	avg_late_30d numeric
1	Excellent (750-900)	571	787	0.270	0.39
2	Good (700-749)	822	722	0.319	0.38
3	Fair (650-699)	1164	674	0.349	0.43
4	Poor (580-649)	1440	618	0.393	2.22
5	Very Poor (300-579)	1003	526	0.450	2.22

Almost half of the customers have a poor or very poor credit score. They also show high credit utilization and more late payments which is why credit score is a key risk driver

Loan approval trends: monthly approval rate over time

	month date	applications bigint	approved bigint	approval_rate_pct numeric
1	2022-07-01	326	146	44.8
2	2022-08-01	317	147	46.4
3	2022-09-01	287	114	39.7
4	2022-10-01	342	163	47.7
5	2022-11-01	305	136	44.6
6	2022-12-01	332	140	42.2
7	2023-01-01	357	163	45.7
8	2023-02-01	265	117	44.2
9	2023-03-01	286	142	49.7
10	2023-04-01	341	163	47.8
11	2023-05-01	318	155	48.7
12	2023-06-01	306	146	47.7
13	2023-07-01	294	146	49.7
14	2023-08-01	296	151	51.0
15	2023-09-01	298	151	50.7
16	2023-10-01	354	159	44.9
17	2023-11-01	321	146	45.5
18	2023-12-01	336	144	42.9
19	2024-01-01	309	142	46.0
20	2024-02-01	304	147	48.4
21	2024-03-01	291	140	48.1
22	2024-04-01	302	134	44.4
23	2024-05-01	325	128	39.4
24	2024-06-01	288	83	28.8

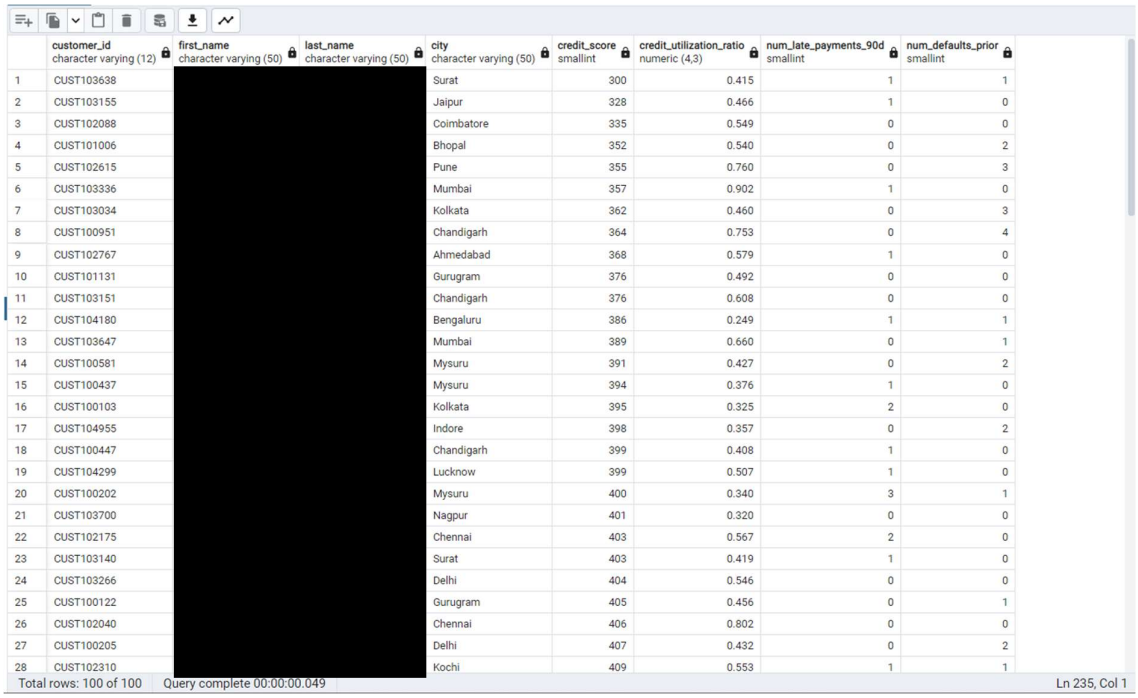
Monthly applications were stable (about 300 per month) and approval rates stayed around 40-51%. The drop in May-June 2024 reflects applications still under review, not a real policy change

Customer segmentation: income band x employment, with approval behaviour

	employment_type character varying (20)	income_band text	customers bigint	avg_credit_score numeric	approval_rate_pct numeric
1	Business Owner	Entry (<6L)	293	623	28.5
2	Business Owner	High (15L+)	51	771	69.4
3	Business Owner	Mid (6-15L)	285	691	57.7
4	Retired	Entry (<6L)	306	596	22.2
5	Retired	High (15L+)	3	720	83.3
6	Retired	Mid (6-15L)	43	693	51.3
7	Salaried	Entry (<6L)	1743	641	34.5
8	Salaried	High (15L+)	93	786	78.8
9	Salaried	Mid (6-15L)	871	718	58.6
10	Self-Employed	Entry (<6L)	619	606	28.8
11	Self-Employed	High (15L+)	45	744	64.0
12	Self-Employed	Mid (6-15L)	376	682	54.3
13	Unemployed	Entry (<6L)	271	495	4.8
14	Unemployed	Mid (6-15L)	1	576	33.3

Approval rates and credit scores rise with income across all employment types. Unemployed customers in the lowest income band have the weakest profile (average score 495, 4.8% approval), while high-income salaried customers have the strongest (average score 786, 78.8% approval).

High-risk customer identification (rule-based watchlist)



	customer_id character varying (12)	first_name character varying (50)	last_name character varying (50)	city character varying (50)	credit_score smallint	credit_utilization_ratio numeric (4,3)	num_late_payments_90d smallint	num_defaults_prior smallint
1	CUST103638	[redacted]		Surat	300	0.415	1	1
2	CUST103155			Jaipur	328	0.466	1	0
3	CUST102088			Coimbatore	335	0.549	0	0
4	CUST101006			Bhopal	352	0.540	0	2
5	CUST102615			Pune	355	0.760	0	3
6	CUST103336			Mumbai	357	0.902	1	0
7	CUST103034			Kolkata	362	0.460	0	3
8	CUST100951			Chandigarh	364	0.753	0	4
9	CUST102767			Ahmedabad	368	0.579	1	0
10	CUST101131			Gurugram	376	0.492	0	0
11	CUST103151			Chandigarh	376	0.608	0	0
12	CUST104180			Bengaluru	386	0.249	1	1
13	CUST103647			Mumbai	389	0.660	0	1
14	CUST100581			Mysuru	391	0.427	0	2
15	CUST100437			Mysuru	394	0.376	1	0
16	CUST100103			Kolkata	395	0.325	2	0
17	CUST104955			Indore	398	0.357	0	2
18	CUST100447			Chandigarh	399	0.408	1	0
19	CUST104299			Lucknow	399	0.507	1	0
20	CUST100202			Mysuru	400	0.340	3	1
21	CUST103700			Nagpur	401	0.320	0	0
22	CUST102175			Chennai	403	0.567	2	0
23	CUST103140			Surat	403	0.419	1	0
24	CUST103266			Delhi	404	0.546	0	0
25	CUST100122			Gurugram	405	0.456	0	1
26	CUST102040			Chennai	406	0.802	0	0
27	CUST100205			Delhi	407	0.432	0	2
28	CUST102310			Kochi	409	0.553	1	1

Total rows: 100 of 100 Query complete 00:00:00.049 Ln 235, Col 1

The watchlist flags customers with a score below 600, utilization above 80%, a prior default, or 2 or more late payments of 90+ days. The 100 lowest-scoring customers have an average score of 422, and more than half have a prior default. **In total, it flags** 1,559 customers (about 31% of the base), mainly due to credit scores below 600. About 39% of flagged customers also have a prior default.

Default rate analysis by loan type (disbursed loans only)

	loan_type character varying (20)	disbursed_loans bigint	defaults bigint	default_rate_pct numeric	avg_interest_rate numeric
1	Home	326	103	31.6	9.2
2	Business	290	86	29.66	15.95
3	Education	336	92	27.38	10.96
4	Gold	386	90	23.32	12.05
5	Auto	670	143	21.34	11.51
6	Personal	1395	283	20.29	17.53

Home loans have the highest default rate (31.6%) while Personal loans have the lowest (20.3%). Interest rates do not follow this risk pattern: Home loans carry the lowest average rate (9.2%), which suggests risk is not fully priced in.

Loan amount distribution by loan type

	loan_type character varying (20)	applications bigint	min_amount numeric	median_amount numeric	avg_amount numeric	max_amount numeric
1	Home	1252	1513000	6879000	6785271	12000000
2	Business	1038	309000	3574000	3622899	6995000
3	Education	727	204000	1976000	2012917	4000000
4	Auto	1375	204000	1317000	1341308	2498000
5	Personal	2500	50000	780000	781944	1500000
6	Gold	608	31000	397000	399229	799000

Loan sizes vary widely by type: Home loans are the largest (median about ₹68.8 lakh) and Gold loans the smallest (median about ₹3.97 lakh). Personal loans receive the most applications (2,500). Loan types with larger amounts, such as Home and Business, also show higher default rates.

Monthly approval statistics with disbursed value

	month text	applications bigint	approved bigint	rejected bigint	disbursed_amount_mn numeric
1	2022-07	326	146	180	208.55
2	2022-08	317	147	170	215.23
3	2022-09	287	114	173	148.07
4	2022-10	342	163	179	193.55
5	2022-11	305	136	169	191.8
6	2022-12	332	140	192	169.57
7	2023-01	357	163	194	195.65
8	2023-02	265	117	148	162.1
9	2023-03	286	142	144	191.94
10	2023-04	341	163	178	242.01
11	2023-05	318	155	163	203.49
12	2023-06	306	146	160	178.56
13	2023-07	294	146	148	170.87
14	2023-08	296	151	145	174.89
15	2023-09	298	151	147	178.38
16	2023-10	354	159	195	213.15
17	2023-11	321	146	175	207.93
18	2023-12	336	144	192	162.57
19	2024-01	309	142	167	153.23
20	2024-02	304	147	157	189.02
21	2024-03	291	140	151	195.7
22	2024-04	302	134	168	163.58
23	2024-05	325	128	133	179.76
24	2024-06	288	83	96	122.64

Monthly disbursed value averaged about ₹184 Mn, totalling ₹441 crore over 24 months. The lower approval rate in May-June 2024 is due to 173 applications still Under Review, not a real decline in approvals.

Risk categorisation blending bureau score and utilisation

	risk_category text	customers bigint	avg_score numeric
1	Low Risk	992	751
2	Medium Risk	2268	674
3	High Risk	1740	558

A rule-based categorisation using credit score and utilization places 19.8% of customers in Low Risk, 45.4% in Medium Risk and 34.8% in High Risk, so a large part of the customers carries high or medium risk.

Transaction behaviour analysis

	customer_id character varying (12)	txn_count bigint	total_inflow numeric	total_outflow numeric	emi_outflow numeric	avg_balance numeric
1	CUST100096	12	695246.80	478836.73	110565.90	629633
2	CUST103209	8	321131.30	433620.92	66776.87	401137
3	CUST102880	9	156114.69	401411.47	123876.61	424218
4	CUST101748	7	0	355253.73	45824.44	546945
5	CUST100058	7	308799.86	349953.86	147367.78	538324
6	CUST104270	11	10504.27	338960.22	58607.85	292001
7	CUST104058	11	258696.30	331661.80	111652.44	291548
8	CUST100484	10	162964.05	317259.81	68055.28	233133
9	CUST104945	12	0	309264.67	103655.95	195336
10	CUST102232	10	0	308338.59	85050.15	194057
11	CUST104975	7	0	308129.04	3076.09	316516
12	CUST103466	10	0	294186.91	77360.23	198782
13	CUST102882	11	254118.17	293107.20	92735.86	273552
14	CUST101240	7	342398.30	293027.10	0	389905
15	CUST103703	10	128749.57	291859.55	0	197885
16	CUST101969	11	41489.48	289838.25	70223.57	351178
17	CUST103990	11	243875.12	289287.60	0	318612
18	CUST101459	12	115008.92	287118.25	108141.47	152864
19	CUST101895	12	219887.22	284888.23	24812.09	191850
20	CUST100575	8	53076.94	279408.32	0	331236
21	CUST103243	9	0	277802.71	65753.26	323963
22	CUST103896	7	0	276420.62	20022.26	288956
23	CUST104911	12	126173.45	276402.21	43632.89	115869
24	CUST101651	11	128815.87	273891.64	69989.79	251561
25	CUST100389	10	55534.82	268232.13	63933.00	303239
26	CUST104681	10	98220.25	267364.53	24128.94	222171
27	CUST102363	11	138027.47	261706.46	0	228436
28	CUST100191	9	58995.08	258366.24	61466.63	257102

Total rows: 50 of 50 Query complete 00:00:00.228

Among the 50 highest-spending customers, 78% had outflows exceeding inflows over 6 months, and about 16% of their spending went to EMI payments. So, inflow, outflow and EMI were used as model features.

Top predictors of loan default

	driver text	default_rate_pct numeric	n bigint
1	credit_score<600	40.11	379
2	utilization>0.8	33.33	36
3	prior_default>=1	33.1	142
4	dti>0.5	28.5	2340

Default rates are highest for customers with credit scores below 600 (40.1% vs 23.4% overall), followed by prior defaults (33.1%). These findings match the model's top predictors, credit score and debt-to-income ratio. Customers with a debt-to-income ratio above 0.5 also default more (28.5%).

7. Feature Engineering

The transactions dataset was aggregated to customer grain using customer_id as the grouping key. For each customer, balance_after was averaged, and transaction type was counted, along with other aggregations to produce 7 summary features.

The 7 summary features created for the models were,

- Average Balance: avg_bal
- Transaction Count: trns_count
- Total Credit: total_credit
- Total Debit: total_debit
- EMI outflow: emi_outflow
- Net Cash Flow: net_cash_flow
- Inflow Outflow Ratio: inflow_outflow_ratio

Two customers had zero total debit, so the inflow-to-outflow ratio was infinite. These were capped at the 99th percentile of the ratio. Customers with no transactions at all were given 0.

These features were made in summary_df and after which duplicate checks were made. After this all datasets were merged using customer_id as the join key.

The cleaned Loan Application dataset was used as a base table to merge customer dataset and cleaned credit history dataset as it held the target variable which is (loan_default) and one row per application. The customer, credit history and summary transaction data were merged using left join to make sure no application records were lost. The shape of the merged dataset was (7500, 56).

As a part of feature engineering, 3 additional features were created as per PRD 13.2 which are

- Outstanding debt to credit limit ratio(debt_to_lmt_ratio)
- Requested amount to annual income ratio(ln_to_incm_ratio)
- Tenure: the number of years between customer_since and application_date. 418 applications were dated before customer_since, so their tenure was set to 0.

The merged dataset was then filtered to rows where disbursed_flag = 1 (loans that were actually disbursed), as per PRD 13.1, leaving 3,403 loans.

Preprocessing

Different categorical columns were turned to numerical using one hot encoding the columns are "employment_type", "residence_type", "loan_type" & "education".

After which leakage control was done where certain column which were not used in Modelling were dropped. The columns approval_status, approved_amount, decision_date and disbursed_flag were dropped because they are only known after the loan decision, so the model would be learning from the answer. Email and phone were also dropped as they are personal data.

Apart from these columns other Text and date columns such as city, state, DOB were not used in the model

Only two columns had empty values, and neither was imputed. loan_default is empty for the 4,097 applications that were never disbursed; filling it would mean inventing an outcome, so these rows were removed by keeping only disbursed loans (PRD 13.3: drop rows with a missing target). decision_date is empty for 173 applications still Under Review; this column was removed for leakage control anyway. After these steps, the model features had no missing values.

Numeric features were standardised for Logistic Regression only, using a scaler fitted on the training data. Tree-based models used the original values.

The final features in the dataset had 54 columns.

8. Baseline Models

Train/Test Split

The total data we have after filtering was 3403. The split was made 80/20 according to the PRD 13.4. 80% of data was used to train the model and 20% of data was used to test the model. 2722 training loans were used to train the model and 681 loans were used to test the model. The split was stratified on loan_default. This keeps the default rate about the same (around 23.4%) in both sets. The random_state 42 was used so that the split is the same every time. The test set had 159 defaulters.

Baseline Models

The initial 2 baseline models that were created were Logistic Regression and Decision Tree they were made as per the instruction mentioned in PRD 18 week 2. Both the models used class weight "balanced" as only 23% of loans default which makes the models pay more attention to defaulters, Logistic Regression is the interpretable baseline and the Decision Tree is a simple non-linear benchmark (PRD 13.5). LR used scaled data where Decision Tree used the original data with max_depth = 5, as a deeper tree might overfit

Baseline Results

Model	ROC-AUC	Recall	Precision	Accuracy	F1
Logistic Regression	0.703	0.610	0.390	0.686	0.475
Decision Tree	0.662	0.642	0.360	0.651	0.462

Results on the test set (681 loans) at the default 0.5 threshold.

Confusion Matrices and observation

Logistic Regression

	Predicted: No Default	Predicted: Default
Actual: No Default	370	152
Actual: Default	62	97

Decision Tree

	Predicted: No Default	Predicted: Default
Actual: No Default	341	181
Actual: Default	57	102

Observation

- LR did better at ranking risks (ROC-AUC 0.703 vs 0.662)
- DT caught slightly more defaulters (102 vs 97), but it also wrongly flagged more good loans (181 vs 152)
- Since neither of the models achieved the target of (ROC-AUC ≥ 0.80, recall ≥ 0.70) stronger models were tried further.

9. Advanced Models & Final Test

Random Forest and XGBoost

Since the baseline models didn't yield the results needed advanced models were created which are Random Forest Model and XGBoost as per PRD 13.5.

Random Forest makes many decision trees and combines their results to give the final prediction. It used class_weight "balanced" just like the baseline models, so that it pays more attention to the defaulters.

XGBoost also uses trees, but it builds them one after the other, where each new tree tries to fix the mistakes of the previous one. Instead of class_weight it used scale_pos_weight = 3, as there are roughly 3 non-defaulters for every defaulter in the data.

Both models were tuned using GridSearchCV with 5-fold cross validation on the training data only, as per PRD 13.4. The settings that gave the best ROC-AUC were picked. The test set was not used during tuning.

The best settings found were:

- Random Forest: n_estimators 500, max_depth 8, min_samples_split 2, min_samples_leaf 6
- XGBoost: n_estimators 200, max_depth 3, learning_rate 0.01, subsample 1.0, colsample_bytree 1.0

Comparison of all 4 models

Models	ROC-AUC	Recall	Precision	Accuracy	F1
Logistic Regression	0.703	0.610	0.390	0.686	0.475
Decision tree	0.662	0.642	0.360	0.651	0.462
Random Forest	0.683	0.560	0.363	0.668	0.441
XGBoost	0.699	0.579	0.368	0.670	0.450

All four models ended up close to each other, with ROC-AUC between 0.66 and 0.70. At the 0.5 threshold none of them reached the recall target of 0.70, so the threshold needed to be adjusted.

Choosing the Threshold

Since none of the models reached recall of 0.70 at 0.5, the threshold was lowered. A lower threshold means the model flags more loans as default, so it catches more defaulters, but it also wrongly flags more good loans.

To choose the threshold without using the test set, 5-fold cross validation was done on the training data only. For each model, the highest threshold that still gave a recall of at least 0.73 was picked (a little above 0.70 to keep a safety margin).

Models	**CV ROC-AUC**	**Threshold**	**Recall**	**Precision**
XGBoost	0.730	0.45	0.735	0.377
Logistic Regression	0.736	0.44	0.743	0.367
Random Forest	0.722	0.43	0.740	0.364
Decision Tree	0.678	0.48	0.734	0.338

At the same recall, XGBoost gave the highest precision, so it raised the fewest false alarms

Final Test and Champion

Final Test Result

XGBoost at a threshold of 0.45 was chosen as the final model. It was then run on the test set (PRD 13.4). During development the test set was checked more than once, which is noted in Limitations.

Metric	Result	PRD Target
ROC-AUC	0.699	>= 0.80
Recall	0.686	>= 0.70
Precision	0.349	None
Accuracy	0.628	None
F1	0.463	None

Confusion Matrix

	Predicted: No Default	Predicted: Default
Actual: No Default	319	203
Actual: Default	50	109

The model caught 109 of the 159 defaulters in the test set and missed 50. It also flagged 203 good loans as risky. Recall came out at 0.686, which is just 3 defaulters short of the 0.70 target (112 were needed).

XGB Winner Model

XGBoost and Logistic Regression had almost the same CV ROC-AUC (0.730 vs 0.736), so the difference is very small. At the same recall, XGBoost had higher precision, meaning fewer good loans were wrongly flagged. Also, PRD 13.6 asks for feature importance and SHAP from XGBoost or Random Forest, which XGBoost provides. Logistic Regression was the runner-up.

Other Experiments

A few more things were tried to see if the model could be improved, but none of them helped, so they were not used in the final model.

- **SMOTE:** SMOTE creates extra synthetic defaulters in the training data to balance the classes. During trial, the test ROC-AUC dropped to 0.671 and recall to 0.642, which was worse than the final model, so it was dropped.
- **Five extra features:** EMI burden, total late payments, debit share of income, a low credit score flag and a prior default flag were added. The CV ROC-AUC did not improve (0.7335 with them vs 0.7346 without), so they were removed.
- **Grouped split by customer:** some customers had more than one loan, so a split was tried where all loans of the same customer stay in the same set. The results were very similar (XGBoost test ROC-AUC 0.711), so the normal stratified split from PRD 13.4 was kept.

10. Feature Importance & SHAP

As per PRD 13.6, the feature importance from the final XGBoost model was used to find which factors drive default the most. The top 5 were:

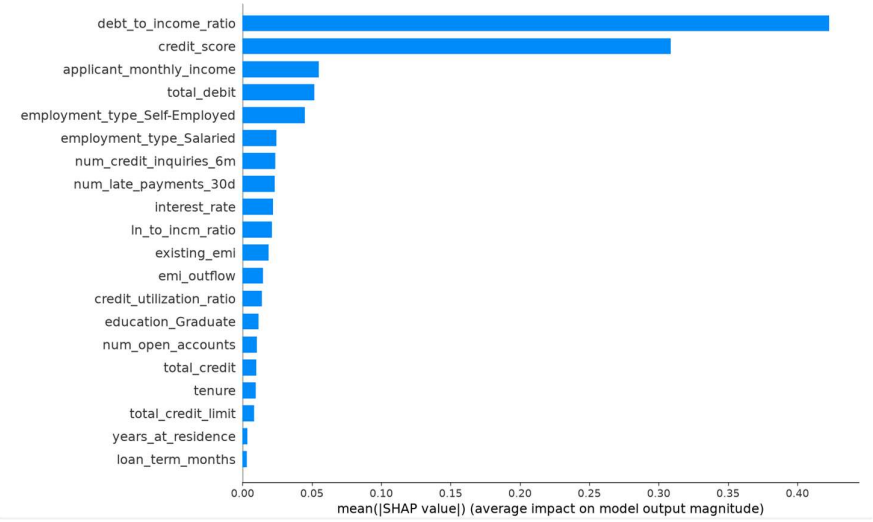
Rank	Features	Importance
1	debt_to_income_ratio	0.105
2	credit_score	0.076
3	applicant_monthly_income	0.059
4	employment_type_Self-Employed	0.038
5	total_debit	0.037

The full ranked list is saved in models/feature_importance.csv.

SHAP Analysis

SHAP values were also used, as per PRD 13.6, to see how much each feature affects the prediction. The chart below shows the average impact of each feature.

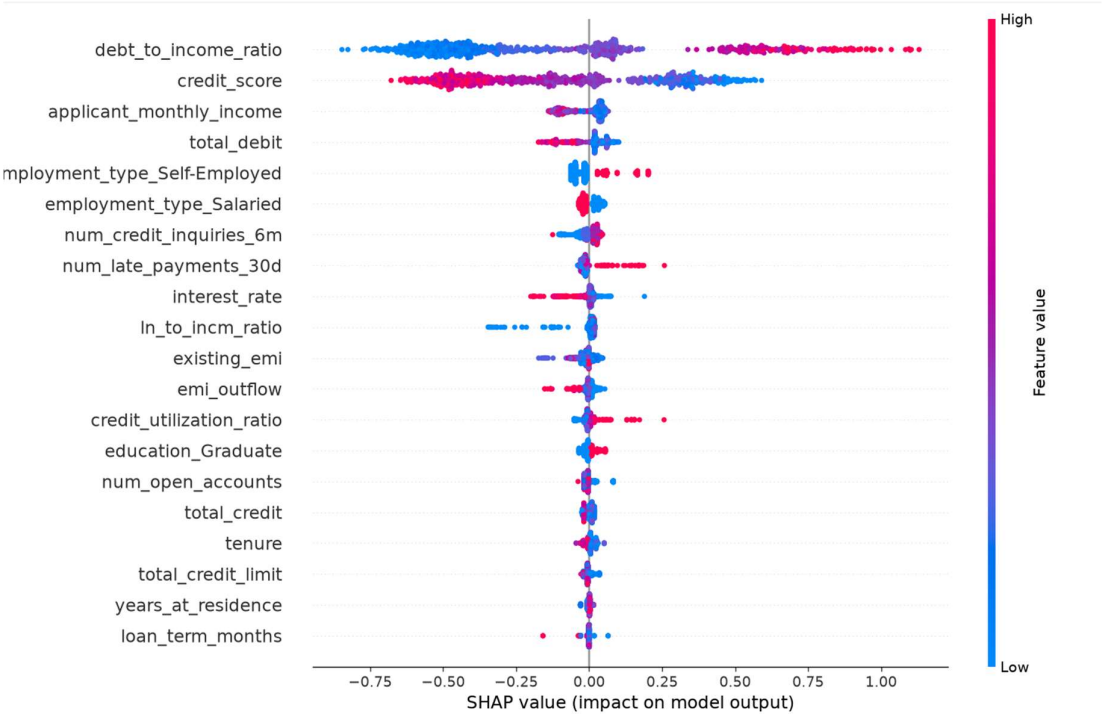
Debt-to-income ratio and credit score are clearly the two strongest drivers, much stronger than the rest. After them come applicant monthly income, total debit and employment type (Self-Employed). This mostly matches the feature importance table above.



SHAP average impact per feature

The second chart shows which way each feature pushes the risk. Each dot is one loan from the test set. Red means a high value of that feature and blue means a low value. Dots on the right side of 0 increase the chance of default, and dots on the left side reduce it.

- **Debt-to-income ratio:** high values (red) are far on the right, so a high DTI increases default risk the most.
- **Credit score:** high scores (red) are on the left and low scores (blue) on the right, so a low credit score increases risk.
- **Monthly income:** higher income pushes slightly to the left, so it lowers risk a little.
- **Employment type:** being Self-Employed increases risk, while being Salaried lowers it.
- **Late payments (30 days) and credit utilization:** higher values push to the right, so they increase risk.
- **Interest rate:** higher rates push to the left, which looks odd at first. This is because the interest rate mostly reflects the loan type. Home loans have the lowest rates but the highest default rate (31.6%), as seen in the SQL results.



SHAP summary plot showing the direction of each features effect

Cross check with SQL results

As per PRD 13.6, the top drivers from the model were cross-checked with the SQL query on top predictors of default (query 7.10). Both agree: customers with a credit score below 600 had a 40.1% default rate, and those with a debt-to-income ratio above 0.5 had 28.5%, and these are also the top 2 drivers in SHAP. Credit utilization above 0.8 (33.3%) and prior defaults (33.1%) also showed high default rates in SQL, but they affect very few loans, so they don't rank high in the model.

Both methods point to debt-to-income ratio and credit score as the main drivers of default.

11. Risk Scoring

As per PRD 13.7, the default probability from the final XGBoost model was turned into a risk score from 0 to 100, using risk_score = round(probability × 100). Customers were then put into three risk bands:

- Low Risk: score 0 to 33
- Medium Risk: score 34 to 66
- High Risk: score 67 to 100

The model was trained on disbursed loans only and then applied to all 7,500 applications to score every customer. Where a customer had more than one application, the highest probability was kept, so each customer has one score. This gave scores for 3,904 customers. The other 1,096 customers had no loan application, so they were not scored.

The results were saved in outputs/scored_customers.csv with customer_id, default_probability, risk_score, risk_category and predicted_default (1 if the probability is 0.45 or more).

Risk Distribution

Risk Band	Customers	Share
Low Risk	529	13.6%
Medium Risk	1423	36.4%
High Risk	1952	50%
Total	3904	100%

Half of the scored customers fall in the High-Risk band. This is partly because XGBoost was trained with scale_pos_weight = 3, which pushes the predicted probabilities up, so more customers cross the score of 67.

Actual Default Rate per Band

To check if the bands make sense, the actual default rate was looked at for customers who had a disbursed loan (2,309 customers):

Risk Band	Customers with Disbursed Loan	Actual Default Rate
Low Risk	465	9.0%
Medium Risk	1014	29.3%
High Risk	830	47.3%

The default rate goes up clearly from Low to Medium to High, so the bands separate good and bad customers well. One thing to note is that these loans include the ones the model was trained on, so the numbers are a little better than they would be on completely new data.

12. Dashboard

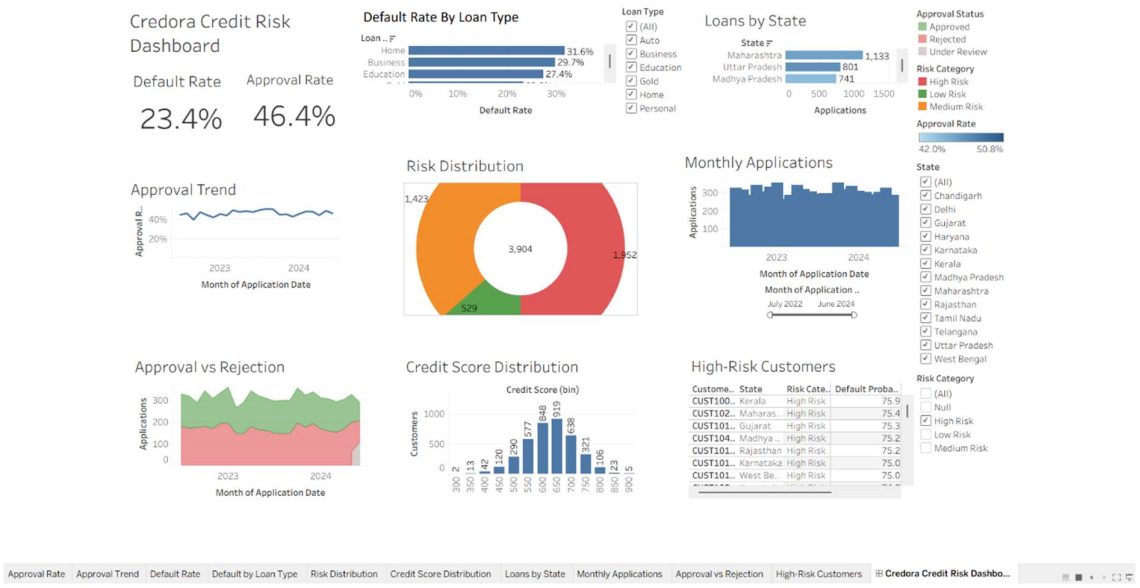
As per the requirements mentioned in PRD 14 created a dashboard in Tableau with the 8 components required

1. Loan approval rate: KPI card (46.4%) and a monthly trend line
2. Default rate: KPI card (23.4%) and a bar chart by loan type
3. Customer risk distribution: donut chart (Low 529, Medium 1,423, High 1,952)
4. Credit score trends: histogram of credit scores
5. Geographic loan distribution: bar chart by state, with approval rate and disbursed value
6. Monthly loan applications: column chart by month
7. Approval vs rejection trends: stacked area chart by month
8. High-risk customer list: table of High-Risk customers, sorted by default probability

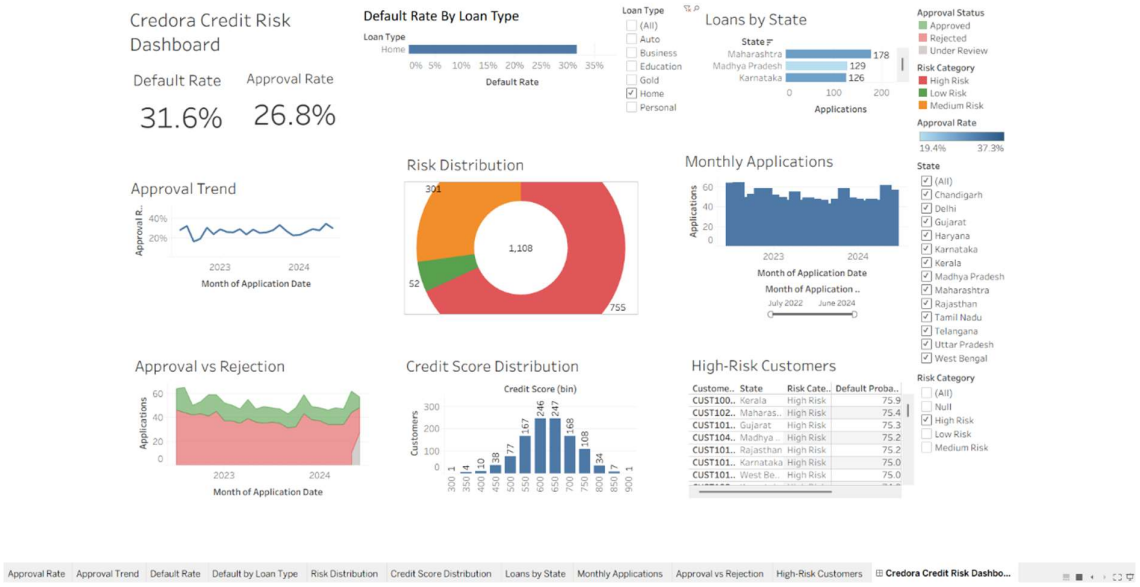
For credit score trends, a histogram was used instead of a line chart, because the credit history data has only one snapshot per customer (all in 2024), so there is no trend over time to show. PRD 14 allows either a line or a histogram.

Filters and Screenshots

The dashboard has the 4 filters suggested in PRD 14: loan type, state, risk category and application month. Loan type, state and month apply to the whole dashboard, so all the charts update together. The risk category filter applies only to the high-risk customer list, so the KPI cards and charts always show the full portfolio. As per FR-10, the high-risk list can be filtered by both risk category and state.



Dashboard Full



Dashboard filtered by home loans only

Dashboard Checks

Before finishing, the dashboard numbers were checked against the PRD and the SQL results:

- Approval rate 46.4% and default rate 23.4% match PRD 1.1.

- Risk distribution (529 / 1,423 / 1,952) matches the scored_customers file.
- Applications total 7,500 across 13 states, and total disbursed value is about ₹441 crore, matching the SQL results.
- The high-risk list shows 1,952 customers when filtered to High Risk.

As per PRD 8, no personal data is shown on the dashboard. Names, email, phone and date of birth were hidden, and only customer IDs appear. The dashboard is saved as dashboard/credit_risk_dashboard.twbx

13. Reproducibility & Project Structure

Reproducibility

As per PRD 8, the whole project can be run with just one command which is python main.py from the project folder. This one command loads the cleaned data, creates the features, trains the models, does the final test, saves the model and also creates the scored customers file.

- The whole run takes around 4 minutes which is under the 5 minute limit given in PRD 8.
- random_state = 42 was used everywhere so that the results are the same every time it is run. To check this the pipeline was run twice and both times the results were the same.
- All the file paths used are relative paths so the project can be run on any computer.
- All the Python packages needed are listed in requirements.txt.

Project Structure

The project folders were made as per the structure given in PRD 15:

- datasets: has the raw and cleaned CSV files. These are not shared as they have personal data of the customers.
- sql: has the database schema and the 10 analytical queries
- notebooks: has the pre-analysis, data cleaning and modelling scripts
- src: has the code for the pipeline (data loading, features, training and risk scoring)
- models: has the saved model, feature importance table, SHAP plot and the model card
- outputs: has the scored_customers.csv file
- dashboard: has the Tableau dashboard and the screenshots
- documentation: has the PRD and the data dictionary
- main.py: runs the whole pipeline
- README.md: explains how to set up and run the project

14. Limitations

As per PRD 16 the target was ROC-AUC ≥ 0.80 and recall ≥ 0.70. The final XGBoost model got ROC-AUC 0.699 and recall 0.686, which is 3 defaulters short of the recall target. All four models stayed between 0.66 and 0.70 on the test set (around 0.73 in CV), so the limit looks like it's in the data and not in the model choice.

As per PRD 13.4 the test set should be used only once at the end. During development it was checked more than once, so the test results may be a little optimistic.

809 customers have 2 or more disbursed loans and they hold 55.9% of all disbursed loans, so the same customer may show up in both train and test. A grouped split was tried as a check and the results were similar.

PRD 17 asks for a score for every customer, but 1,096 customers never applied for a loan, so they have no score. In total 3,904 customers were scored.

The model was trained on disbursed loans only (PRD 13.1) but was used to score all applications. For customers with more than one application, the highest probability was kept.

The default rates for each band include the loans used in training, so they look slightly better than they really are.

About 50% of customers fall in the High-Risk band because of the class weight (scale_pos_weight = 3) used to improve recall.

The UNIQUE constraint for email id had to be removed to keep the data of 2 different individuals because they shared a common email id

The correlation heat map can only be used as a rough guide as it mixes loan and transaction level rows.

Text columns like state, city, occupation and marital status, and the raw date columns, were not used in the model. They may still hold some useful information.

No fairness or bias check was done on the model. This is listed under future work in PRD 19.

15. Conclusion & Future Work

Conclusion

As per the objectives in PRD 3, this project covered the full flow from the raw data to a working dashboard. The 4 datasets were first cleaned and loaded into PostgreSQL, and the 10 analytical queries given in PRD 12.4 were run on them. EDA was done to get a better understanding of the data before moving to the modelling part.

Four models were trained and compared, and XGBoost was picked as the final model with a threshold of 0.45. SHAP was then used to see which features had the most effect on default risk. Using this model, 3,904 customers were given a risk score from 0 to 100 and put into Low, Medium or High risk bands. All of this was brought together in a Tableau dashboard with the 8 components listed in PRD 14. The whole project can be run with just one command and takes about 4 minutes.

The model came close to the PRD 16 targets but did not meet them, with a ROC-AUC of 0.699 and recall of 0.686. Since all four models gave similar results, the data looks like the main limit, so more or better data would help more than trying a different model.

Future Work

As per PRD 19, there are a few things that can be added to this project in the future. A real-time scoring API could be built using FastAPI or Flask so that new applications can be scored straight away during underwriting. An automated retraining pipeline with model drift monitoring would keep the model up to date as the data changes over time. A fair lending and bias review across gender, age and geography should also be done to make sure the model treats all customers fairly.

The model could be improved further by adding more data, such as a fuller transaction history and bureau data over time instead of a single snapshot. A champion–challenger setup with A/B testing would let new models be compared with the current one before replacing it. Finally, a collections prioritisation model could be built on top of the risk score to help the team decide which accounts to follow up on first.